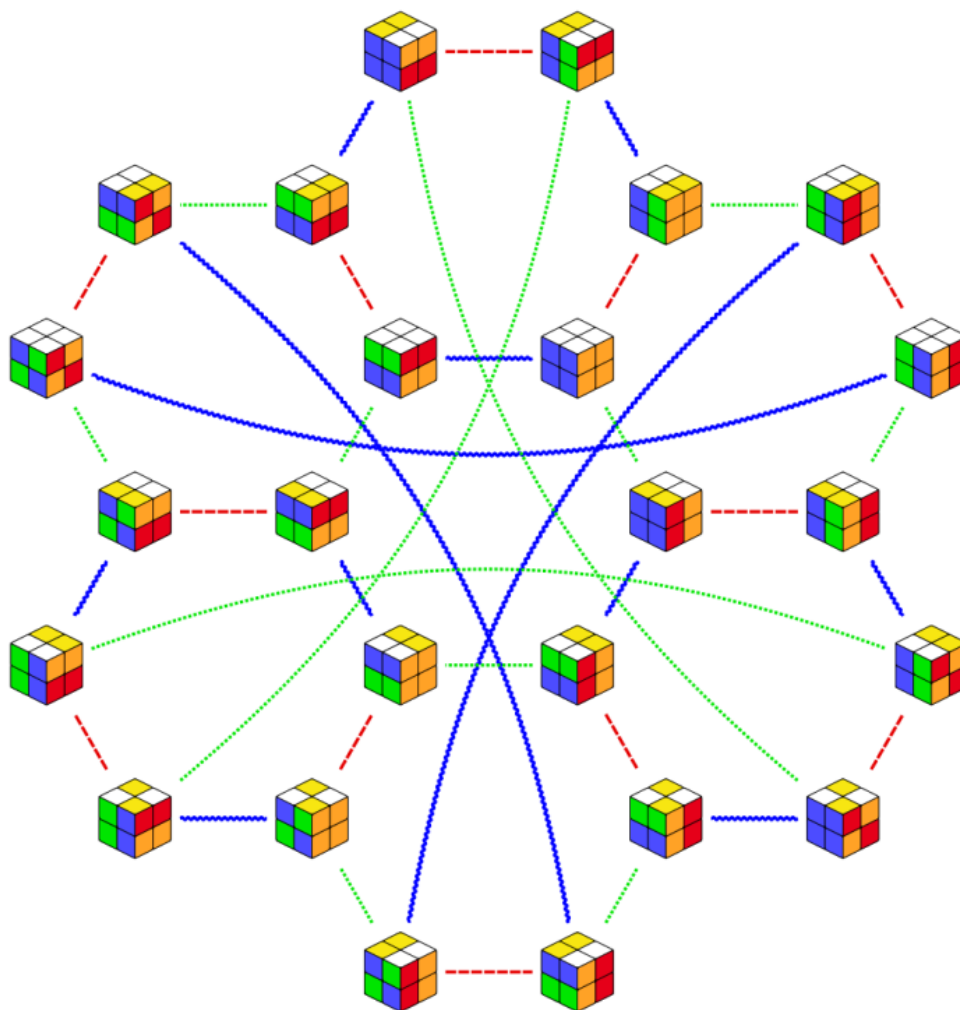


Mu Alpha Theta's Math Journal

Brought to you by: Advait Kamath, Melanie Ma, Derrick Xianto, and MAΘ Board

Special guest writers: Daniel Ma and Sam Stone

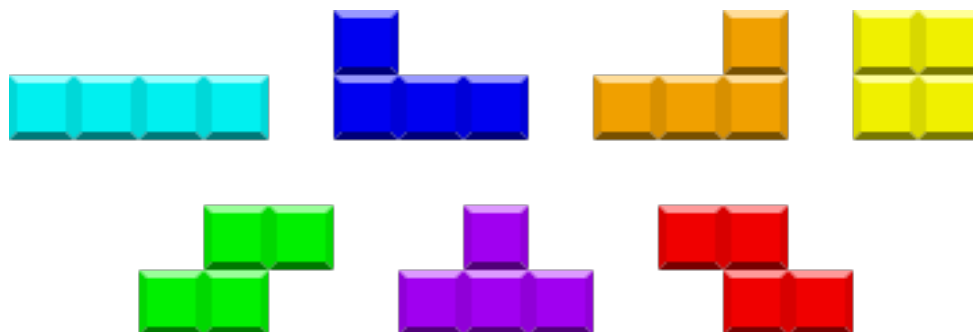


POTM #1 Tetris Time

Email your proofs to mma@imsa.edu for a chance to get on the POTM leaderboard! We will take into consideration quality of work and the order of receipt.

Problem

Tetris is a classic and beloved game by MAO! Modern Tetris uses a **7-bag randomizer**. Each bag contains exactly one of each tetromino (e.g., I, J, L, O, S, T, and Z) arranged in a uniformly random order. After one bag is emptied, a new bag is shuffled independently of all previous bags.



An **I-piece drought** is the number of non-I pieces appearing between two consecutive I-pieces.

Example. In the queue shown to the right, the two cyan I-pieces have three non-I pieces between them, that being a J, an L, and an O. Therefore, this is an I-piece drought of length 3.

For a fixed positive integer n , let S_n be the total length of n consecutive I-piece droughts. How many distinct values can S_n have the value of? Prove your answer.

Proofs can be typed in LaTeX, handwritten, or really whatever format you prefer, as long as it's legible.

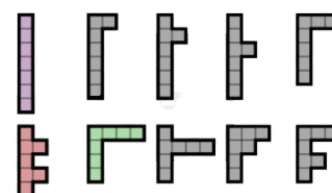


An I-piece drought of length 3.

Bonus: A **heptomino** is a shape formed from seven congruent squares joined edge-to-edge. There are 108 distinct heptominoes up to rotation and reflection. Note that $108 \cdot 7 = 756 = 27 \cdot 28$.

Can one copy of each heptomino tile a 27×28 rectangle?

You may rotate and reflect the pieces.



Examples of heptominoes.

The arts and the math

Creative ways to explain mathematical ideas

An Ode to the Stutterer

You guys have probably seen the quadratic formula,

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

And I'm guessing most of you have some sort of song or mnemonic to remember it. One I've personally heard is sung to the tune of "Pop Goes the Weasel":

"x equals negative b, plus or minus the square root of b squared minus 4ac, all over 2a."

Naturally, after this, you would think, "What if you have a degree 3 polynomial?" Is there a formula to solve these in general? Well, for a while they didn't have one. Famously, Luca Pacioli, who was a collaborator with Da Vinci and helped create the field of accounting, claimed the solution was impossible; fortunately, he was wrong! So, what does this formula look like? Well, given a cubic in the form of $ax^3 + bx^2 + cx + d$, we get the formula:

$$x = \sqrt[3]{\left(\frac{bc}{6a^2} - \frac{d}{2a} - \frac{b^3}{27a^3}\right)} + \sqrt{\left(\frac{bc}{6a^2} - \frac{d}{2a} - \frac{b^3}{27a^3}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3} \\ + \sqrt[3]{\left(\frac{bc}{6a^2} - \frac{d}{2a} - \frac{b^3}{27a^3}\right)} - \sqrt{\left(\frac{bc}{6a^2} - \frac{d}{2a} - \frac{b^3}{27a^3}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3} - \frac{b}{3a}$$

Try making a song out of that!

Well, the song doesn't really help you understand how it was made. Instead of giving you the full, lengthy, algebraic derivation, let's actually explore how they even came up with this formula:

It all started with a little boy, with a slash to his face, on the ground of Brescia, Italy, in 1512. Nicolò Fontana was born to a poor mail rider father in Brescia, Italy. His father died when he was 6, sending his family into even worse poverty (this guy cannot catch a break). After receiving the cut on his face, he was given the nickname Tartaglia. He started studying mathematics at the age of 15. Due to his family's conditions, he had to write on gravestones to get better, and through sheer willpower, he learned enough to gain a position in a university at 18.

He eventually moved to Venice to continue his career. There, some brat challenged him for the spot he earned. They exchanged problems, and the brat's problem set was all depressed cubic problems (a cubic in the form of $x^3 + bx + c$ or a cubic that's sad). Apparently, Scipione del Ferro, one of the greatest mathematicians alive at the time and the teacher of the brat, found a solution to the depressed cubic and hid it from the world.

Tartaglia, who didn't want to lose everything he worked so hard for, decided that he was going to find a way to solve it, and that he did! His solution is as follows:

*Quando chel cubo con le cose appresso
Se agguaglia à qualche numero discreto
Trouan dui altri differenti in esso.*

*Dapoi terrai questo per consueto
Che'llor prodotto sempre sia eguale
Alterzo cubo delle cose neto,*

*El residuo poi suo generale
Delli lor lati cubi ben sottratti
Varra la tua cosa principale.*

*In el secondo de cotestiatti
Quando che'l cubo restasse lui solo
Tu osseruarai quest'altri contratti,*

*Del numer farai due tal part'à uolo
Che l'una in l'altra si produca schietto
El terzo cubo delle cose in stolo*

*Delle qual poi, per communprecetto
Torrai li lati cubi insieme gionti
Et cotal somma sara il tuo concetto.*

*El terzo poi de questi nostri conti
Se solue col secondo se ben guardi
Che per natura son quasi congiunti.*

*Questi trouai, e non con paßi tardi
Nel mille cinquecentè, quatroe trenta
Con fondamenti ben sald'è gagliardi
Nella citta dal mar'intorno centa.*

I'm guessing most of you reading don't understand Italian. But in algebraic terms (if you have $x^3 + px + q$), he set $x = u + v$, then he expanded and, by matching like terms, found that $u^3 + v^3 = -q$ and $u^3v^3 = -p^3/27$. These equations make a quadratic whose roots are u^3, v^3 . Then just add u, v to get x . Shockingly enough, he figured out how to solve it and expressed his solution entirely through poetry! So he kept his position, and he also kept this formula a secret, making sure no one else challenged him.

Cardano, a polymath at the time, begged Tartaglia for the solution, and Tartaglia eventually gave in, under the promise of secrecy. Cardano managed to solve the general case of $ax^3 + bx^2 + cx + d$ by first dividing by a to get $x^3 + (b/a)x^2 + (c/a)x + d/a$, then doing the substitution of $x = y - b/3a$, which if you plug it in, gets the depressed cubic, which Tartaglia has already solved! Cardano then found a way to circumvent Tartaglia's solution, which Cardano published.

Tartaglia's life was pretty rough; even near the climax of his life, he got his work taken advantage of. There is some beauty in it, as he managed to write a poem to solve an unsolved problem while being called Tartaglia (which is a derivation from the word "to stutter" in Italian). As a way to honor his memory, here is a adapted version of his poem that actually has a poem structure in English:

*A cube whose side lengths are d
Combined with a ratio
Equals some b .*

*To solve this equality
Let d be the sum of a and c
Then expand this state completely,*

*We find that this ratio
Equals the product of a and c
Times a constant of negative three.*

*We find that adding 2 cubes
Sides with a and c
these volumes,
Must be exactly b ,*

*This creates a system of pairs
Which equals the unknown cubes
That can be solved with the method of squares*

*Two answers, a conjugate pair
Then take the roots, the radical after the square
Each expression solves a and c 's affair*

*Go all the way to the start
And sum a and c 's part
The question of d must depart.*

*An ode to his creation
Made in distress
An ode to his solution
Now he rests.*

MAO Alumni Spotlight

Daniel Ma, IMSA Class of 2023

Applied Mathematics at Brown University and Quantitative Trader at Citadel



When you were at IMSA, what first got you interested in math, and were there any classes, teachers, or activities that particularly influenced you? At that point, did you already think you would pursue applied math and quantitative finance?

I had participated in many math-related activities before going to IMSA, so I knew I wanted to continue doing so, but it was really the people I met that made me much more interested in math. Having a support group of peers that all came from similar math backgrounds was a hallmark of my time at IMSA. Being able to not only discuss and work through interesting topics but also be in the same courses was an incredible experience. I participated in SIR and off-campus research working in economics and had always enjoyed finance, so applying my mathematical knowledge there was a natural extension. As a result I also found the world of quantitative finance as that perfect intersection and became more interested in pursuing it in my latter years at IMSA.

Why did you choose to study applied math at Brown? How has your experience with math in college differed from your experience at IMSA, and is there a course or topic that has especially changed the way you think about math?

For similar reasons as above, I applied to colleges mostly looking to pursue an applied mathematics major. Initially I almost enrolled in econ, but thankfully Brown's course offerings have many interesting topics applying math to finance. Coming here I've had the privilege of taking courses on Monte Carlo methods, Stochastic Calculus, and more. The experiences here differed in that there's a stronger emphasis on projects and real-world applications in Brown's open-curriculum style of teaching, which I found to be particularly useful once I started thinking about what I wanted my career's work to be like. It made me think differently about how math could extend beyond something I did for "fun" (competing in math competitions) to narrowing down what types of math I'd like to do full-time.

A lot of students have heard the term “quant” but may not know what the job actually looks like. How would you describe your work at Citadel, and what kinds of math and programming do you use on a typical project?

Quant encompasses a huge world of definitions, some particularly bespoke and disjoint from the rest, so it’s probably easier to define it as employing math and statistics-driven techniques to assess and analyze the financial markets. Work can look like researching a particular problem, generating experiments and fine-tuning a model to form targets that can later be translated into trading strategies. It can also look like interacting with brokers across various financial institutions, looking to transact an asset with them and getting the best price for yourself. Each trading firm may have tens to hundreds of such teams/strategies. My work involves math and programming to take on larger-scale projects, but it often requires a lot of fast-paced decision making. We play a lot of board games specifically to train and test the game-theoretic aspect of quant trading, which is incredibly important.

What advice would you give IMSA students who are interested in applied math or quantitative finance? Are there any skills you think they should start building now, or anything you wish you had done differently while you were at IMSA?

It’s never too early! The world of opportunity in this space is huge and so many skills are able to permute into a potential fit, so I wouldn’t worry about fine-tuning or tailoring specific areas to quant. At IMSA I do wish I was more of an architect of my time – I think finding a niche that you’re truly passionate about and have spent a lot of time thinking about, whether that be via research or personal project work, is incredibly rewarding in college. Don’t be afraid to reach out to people or companies whose work interests you, and try to always be curious! I like to say you should always be “adding tools to the toolbox,” which is a mantra I try to follow every day.

Any interesting math projects you have been working on recently that you’re willing to share?

Currently I’m in a class focusing on formal proof and verification, employing the increasingly popular Lean system to build trustworthy programs and fully specified reliable mathematics. Formalization is becoming the cornerstone of this new AI era so I think it’s really important to know what goes into the growing soup of provably valid innovation. Feel free to reach out to me with any questions or if you’d like to chat! My email is dma2459@gmail.com.

MAO Student Spotlight

Sam Stone, IMSA Class of 2029

Many people know the formula for the sum of the first n integers, being $\frac{n(n+1)}{2}$. But what if you keep summing? What's the sum of the first n values for $\frac{n(n+1)}{2}$? What about the sum of that? If we define a sequence of functions $f_0, \dots, f_k$ where $f_0(n) = 1$, and $f_k(n+1) = f_k(n) + f_{k-1}(n+1)$, it's easy to see with this recursion how it equals the sum of the previous function in the sequence. Iterating n all the way up to any given value allows the function to add all values of the previous function. For the first values we get $f_k = 1, n, \frac{n(n+1)}{2}$. Now how would we find f_3 ? By the equation above we have $f_3(n+1) - f_3(n) = \frac{(n+1)(n+2)}{2}$. Now, since the differences between successive values are quadratic, we could guess that f_3 is a cubic function. $f_3(n) = an^3 + bn^2 + cn + d$. By the binomial theorem $f_3(n+1) = an^3 + 3an^2 + 3an + a + bn^2 + 2bn + b + cn + c + d$. Simplifying, $f_3(n+1) = an^3 + (3a+b)n^2 + (3a+2b+c)n + a+b+c+d$. Substituting and simplifying gets

$$f_3(n+1) - f_3(n) = 3an^2 + (3a+2b)n + a+b+c = \frac{(n+1)(n+2)}{2} = \frac{n^2}{2} + \frac{3}{2}n + 1$$

this gives a system of three equations in three variables: $3a = \frac{1}{2}$, $3a+2b = \frac{3}{2}$, $a+b+c = 1$. Solving gives $a = \frac{1}{6}$, $b = \frac{1}{2}$, $c = \frac{1}{3}$. To find d we could test $f_3(1)$ which equals 1 and plugging in the values we got from our formula we get $1 = \frac{1}{6} + \frac{1}{2} + \frac{1}{3} + d$ solving we find $d = 0$. So we have found the solution to f_3 . Earlier we did assume f_3 was a cubic polynomial but there is only one solution for each f_k so we know it is unique. To simplify our solution to $f_3 = \frac{n^3}{6} + \frac{n^2}{2} + \frac{n}{3}$ we can factor out a $\frac{1}{6}$ getting $\frac{n^3+3n^2+2n}{6}$, factoring this expression gives $f_3(n) = \frac{n(n+1)(n+2)}{6}$. Now looking at the previous functions in the sequence it seems to follow the pattern: $\frac{n(n+1)\dots(n+k-1)}{k!}$. How would we prove this? We could try proving by induction.

Proof. The base case being $k = 1$ does satisfy the equality because $n+1 = (n) + 1$ and is in the form $\frac{n}{1!}$. Now let's assume that for all $k \leq a$, $f_k(n) = \frac{n(n+1)\dots(n+k-1)}{k!}$. If

$$\frac{(n+1)(n+2)\dots(n+a+1)}{(a+1)!} = \frac{n(n+1)\dots(n+a)}{(a+1)!} + \frac{(n+1)(n+2)\dots(n+a)}{a!}$$

Then $f_{a+1} = \frac{n(n+1)\dots(n+a)}{(a+1)!}$.

$$\frac{(n+1)(n+2)\dots(n+a+1)}{(a+1)!}$$

By the distributive property this equals

$$\frac{n \cdot (n+1)\dots(n+a)}{(a+1)!} + \frac{(a+1)(n+1)(n+2)\dots(n+a)}{(a+1)!}$$

or

$$\frac{n(n+1)\dots(n+a)}{(a+1)!} + \frac{(n+1)(n+2)\dots(n+a)}{a!}$$

Showing $f_{a+1}(n) = \frac{n(n+1)\dots(n+a)}{(a+1)!}$. So by the inductive hypothesis $f_k(n) = \frac{n(n+1)\dots(n+k-1)}{k!}$. □

So we have solved the problem but there is one cool and seemingly random property that these functions have:

$$\sum_{n=1}^{\infty} \frac{f_k(n)}{2^n} = 2^k.$$

Proof. This bizarre fact seemingly comes out of nowhere but it is easy to see why if we use the definition of the function rather than the actual values. To prove by induction our base case will be $k = 0$. This gives

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = 2^0$$

which is well known to be true. Now assume that

$$\sum_{n=1}^{\infty} \frac{f_k(n)}{2^n} = 2^k$$

for

$$k \leq a$$

Based on the definition of

$$f_k(n)$$

we have

$$\sum_{n=1}^{\infty} \frac{f_{a+1}(n)}{2^n} = \sum_{n=1}^{\infty} \frac{\sum_{i=0}^{n-1} f_a(n-i)}{2^n}.$$

Summing over each fixed i value this yields

$$\sum_{i=0}^{\infty} \sum_{n=1}^{\infty} \frac{f_a(n-i)}{2^{n+i}}.$$

Since

$$\frac{1}{2^i}$$

is a constant for the second sum we can take it out giving

$$\sum_{i=0}^{\infty} \frac{1}{2^i} \sum_{n=1}^{\infty} \frac{f_a(n-i)}{2^n}.$$

Because of our inductive hypothesis this equals

$$\sum_{i=0}^{\infty} \frac{2^a}{2^i}.$$

Which equals

$$2^{a+1}$$

so

$$\sum_{n=1}^{\infty} \frac{f_{a+1}(n)}{2^n} = 2^{a+1}.$$

Hence, we have shown that for all

$$k \geq 0, \sum_{n=1}^{\infty} \frac{f_k(n)}{2^n} = 2^k.$$

□

Student Spotlight Submissions

Open to all IMSA students!

Tell us about something spectacularly interesting in math. It could be anything—an elegant proof, an original problem, a paradox, a poem, a drawing, or another math-related idea you think would be worthwhile sharing!

You can choose to scan the QR code or access the link below to find the submission Google form. You may type your submission in the form or provide a Google Drive link. If you submit a link, make sure its sharing settings are set to “Anyone with the link” and “Viewer.” Instructions and details are also located inside the form.

Here is the link to the form: <https://forms.gle/d4naM8A6VEAd5yAD9>

The hard deadline is **October 30th**. The tentative date for the official MAO Math Journal to be published is **October 31st**. We will select **1–3 submissions** for the Math Student Spotlight section.

If you have any questions or concerns, please email mma@imsa.edu



Scan to submit!

What's new?

New discoveries in mathematics

Around 2 months ago, four mathematicians were announced as the Fields Medal winners for their work in mathematics. The Fields Medal is the most prestigious award a mathematician can receive, with 2 – 4 winners being announced every 4 years. This year, the winners were presented at the International Congress of Mathematicians in Philadelphia, which was the first US-held conference since 1986. The four 2026 winners are Hong Wang, Yu Deng, John Pardon, and Jacob Tsimerman. Additionally, the puzzle answers will be at the end of this article.

Hong Wang and the Kakeya Conjecture

The Kakeya Conjecture originated in 1917, when Sōichi Kakeya asked a question along these lines: What is the smallest area a needle can sweep out while turning to point in every direction? We could spin the needle around its middle, making a full disk, but shifting it as it turns covers less space. So what's the most optimal way the needle can move? A few years after Kakeya's original question, Abram Besicovitch found that a set of line segments pointing in every direction can take up zero area, answering Kakeya's question in 2 dimensions. However, the question remained unanswered in 3 dimensions. This brings us to the conjecture that Hong Wang and her colleague Joshua Zahl proved. They proved the Kakeya Conjecture, which states that needles pointing in every direction in 3D space must still form a fully three-dimensional set, not a thinner fractal. Wang and Zahl posted their 127-page proof in February 2025, earning Wang the Fields Medal. However, the question remains open in four or more dimensions.

Puzzle 1: At age 6, Wang solved this puzzle faster than her father, a math teacher. How can you plant seven trees to get as many straight rows of three trees as possible?

John Pardon: Tangled Knots and Counting Curves

Pick two points on a knotted loop of rope. An ant walking along the rope travels farther than a crow flying straight between them, and a knot's "distortion" is the biggest possible ratio of those two distances. In 1983, mathematician Mikhail Gromov asked whether a certain family of knots, drawn on the surface of a donut, could always be rearranged to keep their distortion below 100. John Pardon proved the answer is no, since some of these knots can be made as tangled as you like. The paper was published in the *Annals of Mathematics*, the field's top journal, while Pardon was still an undergraduate at Princeton. Knot theory is just one of many areas in which Pardon was recognized for the Fields Medal. However, he was honored mostly for his work in symplectic geometry, the math of "phase space," where each point records both where something is and how it's moving. Mathematicians in this field study shapes by counting curves inside them, and Pardon developed new ways to make those counts rigorous. In 2023, he used these ideas to prove the 20-year-old MNOP Conjecture, which says two very different ways of counting curves on certain six-dimensional shapes give the same answer.

Yu Deng: Bouncing Particles and Flowing Gases

At the 1900 International Congress, David Hilbert posed 23 famous problems. However, the sixth asked mathematicians to build the laws of physics from rigorous mathematics. This is the problem that Yu Deng, along with Zaher Hani and Xiao Ma tackled, eventually earning Deng a Fields Medal. Imagine a gas as trillions of tiny billiard balls bouncing off each other. Physicists describe the gas's overall behavior with a simpler equation, the Boltzmann equation. In a 200-page paper, which was reportedly so dense that Deng flew out to Paris to explain his work to other experts, Deng and his collaborators proved that the Boltzmann equation really does follow from the collisions of the individual balls, even over long periods of time. It's important to note that Deng's path as a mathematician was not smooth. As a middle school student in China,

he nearly became a professional Go player, but lost all three final games of the qualifying tournament and thus switched his focus to math. He then failed his first attempt to qualify for China's International Math Olympiad team and missed the final team the next year. Through perseverance, Deng was able to win gold the third time.

Puzzle 2: At age 7, Deng solved a puzzle from his father. Take a square grid that is 2^n cells wide (such as 4×4 or 8×8) and remove any one cell. Prove that the rest can always be covered by L-shaped pieces made of three cells.

Jacob Tsimerman: 99% of Gamblers Quit Before They Hit It Big

If the graph of an equation passes through a few points with whole-number coordinates, that could be luck. But if it passes through many of them, mathematicians had suspected something is forcing it to. The André–Oort Conjecture makes that intuition precise in much more complicated, higher-dimensional spaces, and in 2021, Jacob Tsimerman, Jonathan Pila, and Ananth Shankar were able to prove the conjecture. This work helped earn Tsimerman the Fields Medal. As a professor at the University of Toronto, he is the first medalist based at a Canadian university. Similar to many other Fields Medalists, Tsimerman had a very interesting math path. In seventh grade, he got most of the problems wrong at the Canadian Mathematical Olympiad, where he then spent hours at a university blackboard until he solved one he had missed. He went on to practice Olympiad problems five hours a day, won an International Math Olympiad gold medal at 15, and earned a perfect score the next year. Tsimerman credits competitions with teaching him how to stay with a hard problem for hours, even when you are stuck. Following his Fields Medal award, Jacob has since shifted his research focus toward AI safety and is now working for OpenAI as an AI Safety Researcher.

Puzzle Answers

Seven trees: Plant three trees at the corners of a triangle, three at the midpoints of its sides, and one in the center. That gives six rows of three: the three sides, plus three lines running from each corner through the center to the opposite midpoint.

L-shaped tiles: A 2×2 grid with one cell removed is exactly one L-piece. For a bigger grid, split it into 4 equal quarters. The missing cell sits in one quarter, so place one L-piece at the center of the grid covering one corner cell of each of the other three quarters. Now every quarter is a smaller grid with one cell missing, and you repeat the same step until you reach 2×2 .